Saugat Bhatta

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Education

University of Texas at Arlington — Arlington, TX
Bachelor of Science in Computer Science, (CGPA: 3.73) — *Dec 2026*

Experience

State Farm Insurance, Software Engineer Intern — Jun 2026 – Present
Property and Casualty Modeling — *Richardson, TX*

- Develop and maintain **Python** packages used across 100+ machine learning models, working with data scientists to build modeling and automation tools.
- Build automation tools with **Python**, **Databricks**, **MLflow**, and **AWS** to speed up model development, experiment tracking, and deployment, and used **MLflow** Autologging to finish a planned three-month project in one month.
- Modernized the internal Knowledge Repository by moving its data workflows off **Apache Airflow** into **Terraform** jobs on **AWS** and migrating its search from **Amazon Kendra** to **Amazon Bedrock**, improving reliability and answer quality.
- Building an **MCP** server and a command-line tool (**CLI**) so other engineering teams can use the Knowledge Repository.
- Collaborate in an **Agile** team through daily standups and sprint planning, partnering with engineers and data scientists and owning assigned tasks from planning through delivery.

Undergraduate Researcher, Architecture Optimization for Edge AI — Jan 2026 – Present
University of Texas at Arlington – Advisor: Dr. Chenxi Wang — *Arlington, TX*

- Writing a first-author research paper (**in preparation for publication**) on shrinking a **DeiT-Small** Vision Transformer to run in real time on the **NVIDIA Jetson Orin Nano**.
- Built a repeatable **PyTorch** pipeline that combines model pruning (**PQV-Mobile**), **ONNX** export, and **NVIDIA TensorRT** compilation, measured directly on the device.
- Made the model run up to **4.7× faster** (330 FPS) and **69% smaller** with almost no drop in accuracy, and showed the speedup came from **TensorRT**, not from pruning.

Dallas AI Summer Program — Jun 2025 – Aug 2025
Dallas AI — *Dallas, TX*

- Built a secure full-stack platform using **Spring Boot** and **React** with JWT authentication, enabling real-time AI-powered chat guidance for 100+ simulated users in emotional eating support scenarios
- Researched and implemented ML models with **Hugging Face Transformers** to detect emotional vs. physical hunger patterns, achieving 87% classification accuracy across 1,500+ annotated food-mood records.
- Integrated an **LLM**-powered response engine that turned detected hunger patterns into personalized, real-time coaching messages within the chat interface.

Projects

Distributed Microservices System | *Python, Spring Boot, React, Kafka, Redis, Docker* — Dec 2025 – Present
Distributed event-driven microservices architecture with real-time monitoring and scalable orchestration. — Code

- Implemented **Kafka**-based asynchronous messaging, **Redis** caching, and a **PostgreSQL** persistence layer.
- Configured an **Nginx** API gateway and containerized services with **Docker** for scalable orchestration.

Insurance Smart Policy Advisor | *Python, FastAPI, SQLAlchemy, scikit-learn, SQLite* — Dec 2025 – Present
Full-stack insurance underwriting platform with ML-based risk scoring and premium estimation. — Code

- Engineered structured risk features and trained **scikit-learn** classification models on 1,000+ insurance records.
- Built **FastAPI** prediction APIs with explainable scoring and persisted 200+ generated quotes.

MavThread | *Java, React, Spring Boot, MySQL, JWT, SLF4J, AWS RDS* — Jun 2025 – Oct 2025
Scalable full-stack university discussion platform with secure authentication and high-volume API handling. — Code

- Designed a **Spring Boot** backend deployed on **AWS RDS**, supporting 200+ users and 5K+ API calls.
- Integrated a **React** frontend with structured logging, reducing error resolution time by 40%.

Leadership

LRC Tutors Team Lead (LEAD Team) — Aug 2024 – Present
University of Texas at Arlington — Website

- Lead and coordinate 35+ tutors across scheduling, operations, and performance analytics, and manage the department's **Sitecore** CMS and official mathematics webpages.
- Awarded **Mavs Up MVP** for outstanding cross-department impact and operational excellence.

Technical Skills

Languages: Python, Java, SQL, JavaScript, C/C++

Frameworks & Libraries: PyTorch, TensorFlow, ONNX, TensorRT, FastAPI, Flask, Spring Boot, React, Node.js, MLflow

Distributed Systems & Cloud: AWS, Terraform, Amazon Bedrock, Amazon Kendra, Apache Airflow, Databricks, Kafka, Redis, Nginx, Docker, Google Cloud (Run, Vertex AI, Storage)

Monitoring & DevOps: Prometheus, Grafana, Git